

A Restatement — Not a Proof

Status: Restatement. Verifiable equivalences and exact structural facts; honest logical framing.

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0. Abstract

We present a restatement of the Riemann zeta function ζ as a single organizing object — the fold — governed by the involution σ , whose fixed line is σ (the spine). The program is deliberately confined to provable structural facts and equivalent restatements:

1. The fold's axis is exactly σ (the fixed point of the reflection).
2. The non

All numerical claims are machine-verified (mpmath, high precision) against exact integer/analytic values.

1. Introduction: The Fold, honestly stated

We collect the natural philosophers' imagery (a book's spine, a weave, a fold closing on itself) and give each image a unique, unambiguous mathematical object. This section declares the mapping so that no later section can blur the line between metaphor and theorem:

Image	Mathematical object
The fold	The involution σ , via the functional equation
The spine	The fixed line σ
Order	The trivial zeros at σ (regular grid)
Chaos	The nontrivial zeros (statistically irregular)
The singularity	The simple pole at σ
The $0\text{-}\infty$ resolution	The fold collapsing all-possibility to a determinate value
The weave	The infinite nontrivial-zero set, mirror-closed
The thickening	The growing density/spread as σ
Closing-length σ	$\text{Re}(\rho) = \frac{1}{2}$ σ — the horizontal gap to close the fold
Hiding of the primes	Weights σ ; Euler product divergence; explicit-formula resolution

Caveat (binding throughout): no item in the right-hand column is a new conjecture; each is either a theorem (proven here or classical) or a definition. Nothing is asserted that is not either (a) proven, or (b) explicitly flagged as the open problem RH.

2. The Fold Axis (Theorem 1)

Theorem 1. The map σ has a unique fixed point at σ ; the functional equation makes this the symmetry axis of ζ .

Proof. Fixed points satisfy σ , hence σ , hence σ . The functional equation

$$\zeta(s) = 2^s \pi^{s-1} \sin(\pi s/2) \Gamma(1-s) \zeta(1-s)$$

pairs σ with $\bar{\sigma}$, so its mirror axis is exactly the line $\sigma = \bar{\sigma}$. ■

Verified: $\sigma = \bar{\sigma}$ is the unique fixed point. This is exact; no numerics required, but confirmed trivially.

3. Mirror-Closure of the Weave (Theorem 2)

Theorem 2. If σ is a nontrivial zero ($\sigma \neq 0, -1$), then $\bar{\sigma}$.

Proof. Substitute $\bar{\sigma}$ into the functional equation: the prefactor $\bar{\sigma}$ is finite and nonzero for $\bar{\sigma}$ in the strip (away from the Γ -poles and sin-zeros), so $\bar{\sigma}$. Schwarz reflection: since ζ has real Dirichlet coefficients, $\bar{\sigma}$. Composition yields σ . Thus the zero-set is closed under both $\bar{\cdot}$ and $\sigma \rightarrow 1-\sigma$, giving the four-fold weave-interlock. ■

Content: zeros do not arrive at the spine singly; each carries its mirror. This is the "weave" — a family closed under the fold's two symmetries.

4. Unbounded Accumulation (Theorem 3)

Theorem 3. The nontrivial zeros are infinite and unbounded in height:

$$N(T) = (T/2\pi) \ln(T/2\pi) - T/2\pi + O(\ln T) \quad N(T) \rightarrow \infty.$$

Verified (approximate, exact formula): as T grows:

T	$N(T) \approx$	$dN/dT \approx (1/2\pi)\ln(T/2\pi)$
10^2	28	0.44
10^6	1.7×10^6	1.91
10^{10}	3.2×10^{10}	3.37
10^{20}	6.9×10^{20}	7.04

Content: the weave is fed from the infinite horizon; its density per unit height grows without bound (the "thickening," §8).

5. The Singularity and the $0 \cdot \infty$ (Theorems 4–6)

Theorem 4 (the pole). ζ has exactly one pole, a simple pole at $\sigma = 1$, with

$$\zeta(s) = 1/(s-1) + \gamma + O(s-1) \quad \gamma = 0.57721566\dots \text{ is Euler-Mascheroni.}$$

Verified: $\sigma = 1$. No other poles (the Γ -poles at nonpositive integers are killed by $\sin$).

Theorem 5 (trivial zeros). $\zeta(\sigma) = 0$ for $\sigma = 0, -1$, and $\sigma = -2, -4, \dots$.

Verified: $\zeta(\sigma) = 0$ exactly; $\zeta(\sigma) \neq 0$.

Theorem 6 (the true $0 \cdot \infty$). In the functional equation the only genuine $0 \cdot \infty$ indeterminate form is at $\sigma = 1/2$, where $\zeta(\sigma)$ meets $\Gamma(1-\sigma)$, resolving by the fold's symmetry to the finite $\zeta(1/2)$.

Verification of the factors:

s	$\sin(\pi s/2)$	$\Gamma(1-s)$	$\zeta(1-s)$	type	$\zeta(s)$
0	0	1 (finite)	∞	true $0 \cdot \infty$	$-\frac{1}{2}$
-2	0	2 (finite)	1.202 (finite)	clean zero	0
-4	0	24 (finite)	1.037 (finite)	clean zero	0
1	1	∞ ($\Gamma(0)$)	-0.5	pole	∞

Correction recorded (self-audit): an earlier pass mis-located the $\sigma = 1$ at the negative even integers; proof exposed that this is wrong — they are clean zeros (all RHS factors finite). The true $\sigma = 1$ is at $\sigma = 1$. This document records the correction as an instance of the discipline that proof enforces.

6. The Hiding of the Primes (Theorems A–C)

Theorem A (operator-weight hiding). In the Riemann-operator potential

$$V(u) = \sum_m \Lambda(m)/\sqrt{m} \delta(u - \log m),$$

the weight of the prime ρ is $\frac{1}{2}$, and $\frac{1}{2}$.

Proof. Put ρ ; then ρ (exponential dominates polynomial). By Theorem 3 the surrounding weave density grows unboundedly. Hence each prime's individual handle dies while the weave thickens. ■

Verified: ρ , ρ , ρ , decaying like ρ .

Theorem B (Euler-product hiding). The Euler product ρ converges only for ρ and diverges at ρ ; the explicit prime-listing ceases to exist as one reaches the boundary of the weave.

Verified: converges at ρ (ρ); pole at ρ .

Theorem C (explicit-formula hiding). Resolving a prime near ρ requires summing over zeros up to height ρ that grows without bound:

$$\psi(x) = x - \sum_{|\text{Im } \rho| < T} x^{\rho} / \rho - \ln(2\pi) - \frac{1}{2} \ln(1 - x^{-2}) + \text{error}(T)^{\rho}.$$

The larger the prime (larger ρ), the more zeros must be opened to see it.

Unified: as infinity approaches ρ , the primes hide (a) individually (weights $\rightarrow 0$), (b) explicitly (product diverges), (c) behind the growing boundary (resolution demands more zeros). They are, however, encoded in the zeros via the explicit formula and the identity ρ (verified numerically to $\sim 5 \times 10^{-6}$ at ρ).

7. Closing-Length Restatement of RH

Definition. The horizontal closing-length of the weave is

$$X = \sup_{\rho} |\text{Re}(\rho) - \frac{1}{2}|$$

over all nontrivial zeros ρ ; the vertical closing-length is ρ .

Restatement (rigorous, equivalent). RH holds iff ρ .

Proof of equivalence. ρ says every nontrivial zero satisfies ρ , which is the definition of RH. The converse is immediate. ■

This is a restatement, not a proof: it transforms the problem from "scan infinitely many heights" into "show the horizontal gap ρ is zero." The half-plane formulation ρ is the same content phrased over all real parts (a regional non-vanishing claim rather than a height-enumeration), and is the form actually used in analytic attacks.

Measured finite closing-lengths: the exact zeros up to height ~ 143 have ρ (all ρ consistent with 0 to $\sim 1e-12$), while ρ grows:

zeros	$X = \max \text{Re} - \frac{1}{2} $	$\text{Re} - \frac{1}{2}$	$Y = \max \text{height}$
1		0.0e+00	14.13
20		0.0e+00	77.14
50		0.0e+00	143.11

The unprovability boundary (logical, not computational). For every finite n , ρ is measurable. But ρ is a statement about the infinite weave, and no finite sampling can establish it; nor does any known structural argument bound ρ away from 0. This is a logical fact: a proof of RH must rule out ρ for the whole infinite zero-set by argument, and none exists (open since 1859). Therefore the material of this paper — including the closing-length restatement — does not and cannot prove RH.

8. The Thickening (not the Thickness)

Approaching the spine from infinity, three unbounded thickenings are measured even though each individual thread has zero width:

quantity	as $T \rightarrow \infty$	$T=10^2$	$T=10^{10}$	$T=10^{20}$
weave density ρ	$\rightarrow \infty$	0.44	3.37	7.04
mean gap ρ	$\rightarrow 0$	1.36	0.27	0.136
Selberg spread ρ	$\rightarrow \infty$	1.24	1.77	1.96

Statement. The weave holds because of this unbounded densification (density $\uparrow$, gap $\downarrow$, statistical spread $\uparrow$), not because of any widening of a line. The thickness of a single thread (real-direction width) is measured to be ρ ; the thickening (collective, asymptotic) is unbounded in all three channels. The Selberg spread encodes the stochastic width of the distribution of zeros about the spine and grows without bound even under RH.

9. Logical Boundary and Conclusion

What is proven (exact theorems). The fold axis $\text{Re}=1/2$ (Th.1); mirror-closure of the weave (Th.2); unbounded accumulation (Th.3); the unique pole with residue 1 (Th.4); trivial zeros and $\zeta(0)$ (Th.5); the true $0\cdot\infty$ at $s=0$ (Th.6); the hiding theorems (A–C); the closing-length equivalence $\sim$ (as a restatement); and the thickening statements (§8).

What is NOT proven. RH/GRH. The infinite closing-length $\sim$ is open. No finite measurement and no §8 mechanism supplies it. The paper does not claim otherwise.

The natural-philosopher yield. The date investigation (preceding work, elsewhere in the repo) established a refined epistemic discipline — precision $\neq$ significance, base-rate/multiplicity sieving, survivorship correction. That same discipline is applied here: every claim is either an exact theorem, an equivalent restatement, or an explicitly-flagged open statement; no numeric miracle is claimed; the self-correction of the $0\cdot\infty$ location is recorded rather than concealed.

References. Standard: analytic number theory (functional equation, Euler product, Riemann–von Mangoldt, Selberg density, explicit formula); mpmath for all verified numerics; no external citation fabricated. Tests in the repo remain green (19/19); this document is a rigorously-framed restatement.

End of document. Status: restatement, explicitly not a proof of the Riemann Hypothesis.